ITCS216: Database Management Systems (1/2026)

Faculty of Information and Communication Technology

Workshop on Project Phase 1: Data Modeling

This workshop aims to provide a chance to refine the business rules and develop the data model for Project Phase I. All members of the team are required to analyze the business requirements (both rules and constraints) and sketch the data model using ERD Chen Notations.

Instructions:

1. **Use the group project to do an individual assignment.**
2. Starting from the M1 report, **each member** picks the different business process.
3. Revise / or create at least 3 business rules using **Template#1.**

Template#1: **Format:** Noun + Verb + Noun (e.g., a student enrolls many classes)

Business Process Name: Enrollment		
Rule No. 1	**Rule Statement**	**Relationship**
Forward/Active	A student can enroll many classes.	1:M
Backward/Passive	A class can be enrolled by many students.	1:M
Conclusion	Students can enroll many classes.	M:N

Note: you need to formulate both Backward and Forward

Business Process Name: Class Registration		
Rule No. 1	**Rule Statement**	**Relationship**
Forward/Active	A member can register many classes.	1:M
Backward/Passive	A class can be registered by many members	1:M
Conclusion	Many members can register many classes.	M:N
Rule No. 2	**Rule Statement**	**Relationship**
Forward/Active	A instructor can teach many classes.	1:M
Backward/Passive	A class can only be taught by one instructor	1:1
Conclusion	A instructor can teach many classes.	1:M
Rule No. 3	**Rule Statement**	**Relationship**
Forward/Active	A class can book many room.	1:M
Backward/Passive	A room can be booked for many class.	1:M
Conclusion	Many classes can book many rooms	M:N

4. Revise / or create at least 2 business rules using **Template#2.**

Template#2: **Format**: Noun + Verb + Attributes (e.g., a student comprises of ID, Name, Email),

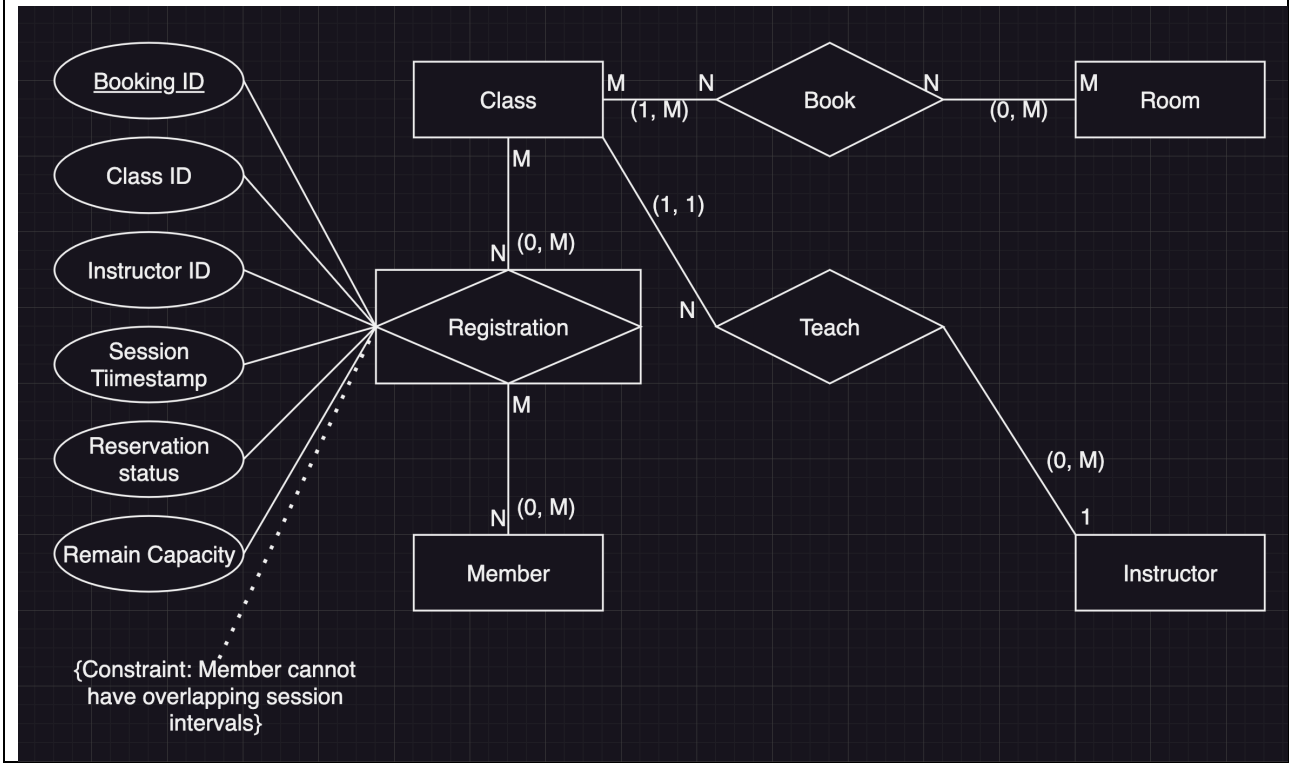
Business Process Name: Enrollment
Rule No. 2
A student comprises of ID, Name, Birthdate, and calculated Age.

Rule No. 8
A class registration must contain Booking ID, Class ID, Instructor ID, scheduled session timestamp, member reservation status, and class remain capacity.
Rule No. 9
A member can only register one class at one time slop, no overlapping allowed.

5. Use the results obtained from Step 2-3 to draw the ERD (CHEN notation) using **Template #3**. All necessary notations for each rule have to demonstrate with complete and correct design

Template #3: Example

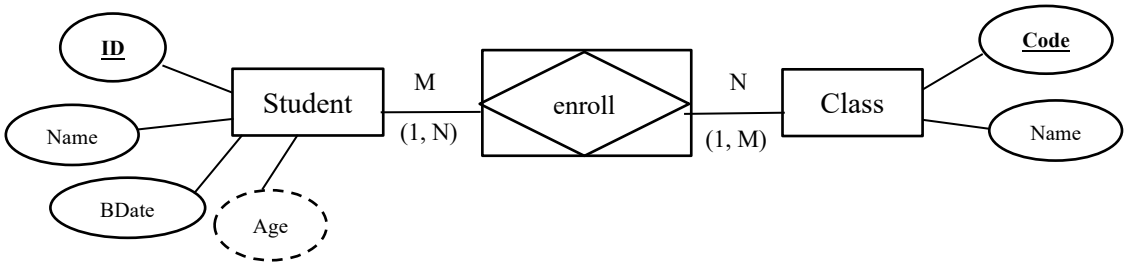
Business Process Name: Enrollment	
Rule No. 1	Many students can register many classes.
Rule No. 2	A instructor can teach many classes.
Rule No. 3	Many classes can book many rooms
Rule No. 4	A class registration must contain Booking ID, Class ID, Instructor ID, scheduled session timestamp, member reservation status, and class remain capacity.
Rule No. 5	A member can only register one class at one time slop, no overlapping allowed.
ERD	



Template #3: Example

Business Process Name: Class Registration	
Rule No. 1	Students can enroll many classes. (M:N)
Rule No. 2	A student comprises of ID, Name, Birthdate, and calculated Age.
Rule No. X	A class consists of class code and name.
...	

ERD



6. Use the result from your teammate to create only one EERD diagram per group.
   (Recommended Tools: https://www.drawio.com/)